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University at Buffalo, The State University of New York, Department of Chemistry, NY

[Google Scholar](#) [ORCID](#)**PROFESSIONAL APPOINTMENTS**

Postdoctoral Associate , University at Buffalo Advisor: Prof. Janet R. Morrow	New York, US Dec. 2021-Present
Postdoctoral Fellow , Sorbonne Université Advisor: Prof. Hélène C. Bertrand	Paris, FR Feb. 2021 – Nov. 2021
Postdoctoral Fellow , Tohoku University Advisor: Prof. Shin Mizukami	Sendai, JP Jul. 2018 – Nov. 2020
Research Scientist , Jubilant Generics Limited	Noida, IN Apr. 2017 – Jul. 2018
Research Internship , CSIR-IMMT	Bhubaneswar, IN May. 2012 – Jul. 2012

EDUCATION

Ph.D. in Chemistry , University of Delhi Advisor: Prof. Satish Kumar Thesis Title: Synthesis and applications of photoactive multifunctional materials	Delhi, IN Nov. 2018 Grade: A
M.Sc. in Organic Chemistry , University of Delhi Thesis Title: Current perspectives of conducting polymers and organic light emitting diodes	Delhi, IN May. 2013 Grade: 1 st Class
B.Sc. in Chemistry (Honors) , Utkal University	Bhubaneswar, IN May. 2011 Grade: 1 st Class

FELLOWSHIPS, AWARDS AND RECOGNITIONS

Postdoctoral Fellowship, University at Buffalo	New York, US 2021-Present
Fondation de la maison de la chimie, CNRS Fellowship	Paris, FR 2021
Featured by C&EN, American Chemical Society (Chemistry in Pictures)	US 2020
Tokyo Biochemical Research Foundation (TBRF) Fellowship	Tokyo, JP 2018 – 2020
SPOT ON award, Jubilant Generics Limited	Noida, IN 2017, 2018
Shortlisted for SciFinder Future Leaders in Chemistry	Columbus, US 2014, 2015
Merit scholarship	Odisha, IN 2007
Meritorious prize	Odisha, IN 2006

PUBLICATIONS

(Total Number of Citations: 590, h-index: 15)

From Postdoctoral Research:

1. **Sahoo P.R.**, Turowski S., Sperryak J., Morrow J. Self-assembled Iron (III) Coordination Cage as an MRI-Active Carrier for a Gold(I) Drug. *Bioconjug. Chem.* (Under review) ([2024](#)).
2. **Sahoo P.R.**, Siriwardana B., Velarde L., Morrow J. Tetrahedral Cages for Formation of Gold(I) Drugs and Protection Against Serum Protein Binding. *Inorg. Chem.* (Submitted) ([2024](#)).
3. Matsui Y., Kowada T., Ding Y., **Sahoo P.R.**, Kikuchi K., Mizukami S. Long-term Imaging of Intranuclear Mg^{2+} Dynamics during Mitosis Using a Localized Fluorescent Probe. *Chem. Commun.* 7048-7051, 59, ([2023](#)).
4. Morrow J., Raymond J.J., Chowdhury M.S.I., **Sahoo P.R.** Redox-Responsive MRI Probes Based on First-Row Transition-Metal Complexes. *Inorg. Chem.* 61, 37, 14487–14499 ([2022](#)).
5. Prioux-Klotz C., Chedotal H., Zoumpoulaki M., Chouzenoux S., Chêne C., Lopez-Sanchez A., Thomas M., **Sahoo P.R.**, Policar C., Batteux F., Bertrand H. C., Nicco C., Coriat R. A New Manganese Superoxide Dismutase Mimetic Improves Oxaliplatin-Induced Neuropathy and Global Tolerance in Mice. *Int. J. Mol. Sci.* 23 (21), 12938, ([2022](#)).

From Doctoral Research:

6. **Sahoo P.R.**, Sairam K., Kumar R., Rana K.P., Kumar S. Synthesis and experimental investigations of a photoactive naphthopyran for sensing nanomolar concentration of ammonia. *J. Photochem. Photobiol. A: Chem.* 115749, ([2024](#)).
7. **Sahoo P.R.**, Kumar N., Sairam K., Gulati L.K., Gulati G.K., Datta A., Kumar S. Tuning fork shaped bisbenzothiazole derivative as a pH responsive digital fluorescent probe and its ex-vivo haemolysis analysis. *Anal. Methods.* 15, 6722-6726 ([2023](#)).
8. Nikam A., **Sahoo P.R.**, Musale S., Pagar R., Paiva-Santos A., Giram P. S. A Systematic Overview of Eudragit® Based Copolymer for Smart Healthcare. *Pharmaceutics.* 15(2), 587, ([2023](#)).
9. Kumar, A.; **Sahoo, P.R.**; Kathuria, I.; Prakash, K.; Kumar, S. Oxazine as an efficient precursor for the development of photochromic spiropyrans. *J. Photochem. Photobiol. A: Chem.* 438, 114541, ([2023](#)).
10. **Sahoo, P.R.**; Kathuria, I.; Kumar, S. Structural arrangement of ligand-metal complex with centered zinc and nickel atoms and their optical features. *J. Mol. Struct.*, 1262, ([2022](#)).
11. Kumar, A.; **Sahoo, P.R.**; Prakash, K.; Arya, Y.; Kumar, S. Light controlled dimerization of spiropyran as a tool to achieve dual responsive capture and release system in aqueous media. *J. Electroanal. Chem.*, 903, 115845, ([2021](#)).
12. Kumar, A; Kumar, A; **Sahoo, P. R**; Kumar, S. A MC-spiropyran for smartphone assisted reversible, selective and nanomolar level detection of formic acid in water and gas phase. *J. Mol. Struct.*, 1223, ([2021](#)).

13. Kumar, A; Sardhalia, V; **Sahoo, P. R**; Kumar, S. Structure analysis and evaluation of two probes for the colorimetric detection of Hg^{2+} and turn-on fluorescence-based detection of Cd^{2+} ions in an aqueous solution. *J. Mol. Struct.* ([2021](#)).
14. **Sahoo, P. R.**, Kumar, A; Kumar, A; Kumar, S. Experimental and computational investigation of polymorphism in methyl 3-hydroxy-4-(piperidin-1-ylmethyl)-2-naphthoate. *J. Mol. Struct.* 1219, ([2020](#)).
15. Kumar, A; Kumar, A; **Sahoo, P. R**; Kumar, S. A light controlled, reversible, sensitive and highly selective colorimetric sensor for mercuric ions in water. *J. Mol. Struct.*, 1206, ([2020](#)).
16. **Sahoo, P. R.**, Kumar, A; Chaudhary, S; Hitaish, Chikkara, R; Malik, K; Sonam; Seth, C; Duggal, G; Kumar, S. Tweezers shaped hydrazone derivatives for the selective detection of cyanide ion. *J. Incl. Phenom. Macrocycl. Chem.*, ([2020](#)).
17. **Sahoo, P. R.**, Kumar, A; Kumar, A; Kumar, S. Synthesis and optical properties of Copper (II) and Nickel (II) complexes of a highly fluorescent morpholine-derivative. *Polyhedron*, 171, 559-570, ([2019](#)).
18. Kumar, A; **Sahoo, P. R**, Arora, P; Kumar, S. A light controlled, sensitive, selective and portable spiropyran based receptor for mercury ions in aqueous solution. *J. Photochem. Photobiol. A: Chem.* 384, 112061, ([2019](#)).
19. Kumar, A; Kumar, A; **Sahoo, P. R**; Kumar, S. Colorimetric and fluorescence-based detection of mercuric ion using a benzothiazolinic spiropyran. *Chemosensor*, 7, 35, 1-19 ([2019](#)).
20. Arora, H; **Sahoo, P. R.**; Kumar, A; Kumar, R; Kumar, S. The direct synthesis of a substituted naphthopentathiepin as selective Co^{2+} ion recognition in aqueous solution. *J. Incl. Phenom. Macrocycl. Chem.*, 1-11, ([2019](#)).
21. **Sahoo, P.R.**; Prakash, K.; Kumar, S. Light controlled receptors for heavy metal ions. *Coord. Chem. Rev.* 357, 18-49, ([2018](#)).
22. **Sahoo, P.R.**; Kumar, S. The experimental and theoretical studies of a merocyanine form based turn off fluorescent sensor for Fe^{3+} ions with nanomolar level sensitivity in aqueous solution. *J. Luminiscence*, 201, 203-210, ([2018](#)).
23. **Sahoo, P.R.**; Prakash, K.; Kumar Ajeet, Kumar, S. Efficient reversible optical sensing of water achieved through the conversion of H-aggregates of a merocyanine salt to J-aggregates. *Chemistry Select*, 2, 5924-5932, ([2017](#)).
24. **Sahoo, P.R.**; Prakash, K.; Kumar, S. Experimental and theoretical investigations of cyanide detection using a photochromic naphthopyran. *Supramol. Chem.*, 29, 183-192, ([2017](#)).
25. Prakash, K.; **Sahoo, P.R.**; Kumar, S. A fast, highly selective and sensitive anion probe stemmed from anthracene-oxazine conjugation with CN^- induced FRET. *Dyes Pigm.*, 143, 393-400, ([2017](#)).
26. Kumar, A.; Prakash, K.; **Sahoo, P.R.**; Kumar, S. Visible light controlled aggregation of a spiropyran- HSO_4^- complex as a strategy for reversible detection in water. *Chemistry Select*, 2, 8247-8252, ([2017](#)).

27. **Sahoo, P.R.**; Prakash, K.; Kumar, S. Synthesis of an Oxadiazole through an indole mediated single step procedure for selective optical recognition of Cu²⁺ ions. *Sens. Actuator B-Chem.*, 242, 299-304, ([2017](#)).
28. **Sahoo, P.R.**; Kumar, S. Synthesis of an optically switchable salicylaldehyde substituted naphthopyran for selective and reversible Cu²⁺ recognition in aqueous solution. *RSC Adv.*, 6, 20145-20154, ([2016](#)).
29. **Sahoo, P.R.**; Kumar, S. Photochromic spirooxazine as highly sensitive and selective probe for optical detection of Fe³⁺ in aqueous solution. *Sens. Actuator B-Chem.*, 226, 548-552, ([2016](#)).
30. **Sahoo, P.R.**; Mishra, P.; Agarwal, P.; Gupta, N.; Kumar, S. Synthesis of tweezer type receptors for colorimetric detection of cyanide ions. *J. Incl. Phenom. Macrocycl. Chem.*, 84, 163-171, ([2016](#)).
31. Prakash, K.; **Sahoo, P.R.**; Kumar, S. A substituted spiropyran for highly sensitive and selective colorimetric detection of cyanide ions. *Sens. Actuator B-Chem.*, 237, 856-864, ([2016](#)).
32. **Sahoo, P.R.**; Prakash, K.; Mishra, P.; Agarwal, P.; Gupta, N.; Kumar, S. Synthesis, X-ray, ¹H-NMR and DFT analysis of the phthalimide-hydrazone probes as selective anion sensor. *Supramol. Chem.*, 29, 417-429, ([2016](#)).
33. Kumar, S.; Mathew, K.M.; Neelam; **Sahoo, P. R.**; Kumar, A. Influence of hydrogen bonding on the conformational stability of calix [4] crown-4-ether. *Indian J. Chem. A*, 54, 52-57, ([2015](#)).
34. Agarwal, P.; Mishra, P.; Gupta, N.; **Sahoo, P.R.**; Kumar, S. (Z)-3-[2-(2, 4-Dinitrophenyl) hydrazin-1-ylidene] isobenzofuran-1(3H)-one dichloromethane hemi solvate. *Acta Crystallogr. E* 70, o418, ([2014](#)).

BOOKS & BOOK CHAPTERS

1. **Sahoo, P. R.**; Saxena A.; Kumar S. Oxadiazole in Material and Medicinal Chemistry. CRC Press, Taylor and Francis Group, **2024**. ([Book](#))
2. **Sahoo, P. R.*** Light Responsive Materials: Properties, Design, and Drug Delivery. ACS Symposium Series Vol. 1436, Chapter 5, 101-127, ACS Books **2023**. ([Book chapter](#))
3. Kumar, S.; **Sahoo, P. R.**; Macwan, V. R.; Kaur, J.; Mukesh, Sahney, R. Macrocyclic Receptors for Environmental and Biosensing Applications. CRC Press, Taylor and Francis Group, **2022**. ([Book](#))
4. **Sahoo, P. R.**; Kowada. T.; Mizukami S. Long-term Mg²⁺ imaging in live cells with a targetable fluorescent probe. *Methods in Molecular Biology (Springer Protocols)* **2021**. ([Book chapter](#))
5. Kumar S.; **Sahoo, P. R.** Ionic liquid based membranes for water softening in Ionic liquid as green solvents (ed.) Elsevier **2019**. ([Book chapter](#))

SELECTED CONFERENCE PRESENTATIONS

1. **Sahoo, P. R.**; Chédotal, H.; Zoumpoulaki, M.; Nicco, C.; Coriat, R.; Policar, C.; Bertrand, H.C. Development of Pt(IV)-SOD mimic conjugates for enhanced anticancer activity with reduced neurotoxicity. FrenchBIC annual meeting held at Obernai, Strasbourg, France 10-13 Oct 2021. (Poster)

2. **Sahoo, P.R.;** Mizukami, S. Super-resolution imaging of Mg^{2+} dynamics in organelles by multifunctional fluorescent probes, TBRF annual meeting, Tokyo 1st March 2019 (Oral)
3. **Sahoo, P.R.;** Neelam, Kumar, A.; Kumar, S. Development of Cu^{2+} selective photoreversible chelator. Light in Chemistry, Materials and Biology (LCMB-2014), an international Conference held at IIT Kharagpur, India 24-25 Feb 2014. (Poster)
4. **Sahoo, P.R.;** Mathew, K. M.; Kumar, S. Photoreversible Chromene derivative for sensing Cu^{2+} ions. (ACP-2014) 8th Asian Photochemistry Conference held at IISER, Trivandrum, Kerala, India 10-13 Nov 2014. (Poster)
5. **Sahoo, P.R.;** Prakash, K; Kumar, S. Colorimetric detection of Cu^{2+} ion using Naphthopyran Scaffold. 10th Mid-Year CRSI Symposium in chemistry held at NIT, Trichy, Tamilnadu, India 23-25 July 2015. (Poster)
6. **Sahoo, P.R.;** Prakash, K; and Kumar, S. Spirooxazine as highly selective probe for optical detection of Fe^{3+} in aqueous solution. "DU-JAIST Indo-Japan Symposium on Chemistry of Functional Molecules/ Materials" February 26-27, 2016 organized by Department of Chemistry, University of Delhi, Delhi 110007, India. (Poster)
7. **Sahoo, P.R.;** Prakash, K; and Kumar, S. A "Turn-On" Optical sensor for selective detection of Fe^{3+} in aqueous media. International conference on Recent advances in chemical sciences March 29-30, 2016 organized by Dept. of chemistry, Aligarh Muslim University, Aligarh, India. (Poster)
8. **Sahoo, P.R.;** Prakash, K; and Kumar, S. A turn-on optical receptor for Fe^{3+} . 19th CRSI National Symposium in chemistry held at University of North Bengal, West Bengal, India 14-16 July 2016. (Poster)

LIST OF MENTORED STUDENTS

1. Kashfia Ahmed, Undergrad (2018-2019), Tohoku University
2. Vaskar Sardhalia, B. Sc. St. Stephen's College (2016-2017): now pursuing Ph.D. at Sorbonne University
3. Ajeet Kumar, M.Sc. Delhi University (2015-2017): now serving as scientist at OLON (API manufacturing company)
4. Harshit Arora, B. Sc. St. Stephen's College (2015-2016): now persuing PhD. at Purdue University
5. Palak Agarwal, B.Sc. St. Stephen's College (2013-2014)

PROFESSIONAL REFERENCES

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